

CHAPTER 7: EQUILIBRIUM (JEE FOCUS)

Topics Covered:

- Chemical equilibrium
- Law of mass action
- Equilibrium constant (K_c , K_p)
- Le Chatelier's principle
- Ionic equilibrium
- pH and pOH
- Buffer solutions
- Solubility product

Core Concepts:

- Equilibrium $\rightarrow$ forward rate = backward rate
- Dynamic nature $\rightarrow$ reactions continue
- $K_c \rightarrow$ concentration based
- $K_p \rightarrow$ pressure based
- $K_p = K_c(RT)^{\Delta n}$
- Le Chatelier $\rightarrow$ system opposes change
- Addition of reactant $\rightarrow$ shift forward
- Increase pressure $\rightarrow$ shift to fewer moles
- Temperature change $\rightarrow$ affects K
- $pH = -\log[H^+]$
- $pOH = -\log[OH^-]$
- $pH + pOH = 14$
- Buffer $\rightarrow$ resists pH change
- Solubility product (K_{sp}) $\rightarrow$ ionic equilibrium constant

Key Formulas:

- $K_c = \frac{[\text{products}]}{[\text{reactants}]}$
- $K_p = K_c(RT)^{\Delta n}$
- $pH = -\log[H^+]$
- $pOH = -\log[OH^-]$
- $pH + pOH = 14$
- K_{sp} = product of ion concentrations
- $K_w = [H^+][OH^-] = 10^{-14}$

Important Patterns:

- $K \gg 1 \rightarrow$ products favored
- $K \ll 1 \rightarrow$ reactants favored
- Only temperature changes K
- Catalyst $\rightarrow$ no effect on K
- Pressure affects only gases
- Common ion $\rightarrow$ decreases solubility
- Buffer $\rightarrow$ weak acid/base + salt
- pH scale $\rightarrow$ logarithmic

Common Mistakes:

- Including solids in K expression

- Wrong Δn in K_p relation
- Forgetting log scale in pH
- Assuming catalyst changes equilibrium
- Ignoring temperature effect on K
- Mistakes in buffer calculations
- Wrong units in equilibrium expressions

Examples (Important Cases):

- Calculate K_c , K_p
- Effect of pressure/temperature
- pH of strong acid/base
- Buffer solution problems
- K_{sp} and solubility comparison

Exam Tips:

- Write equilibrium expression carefully
- Check physical states before including terms
- Use log rules for pH quickly
- Apply Le Chatelier logically
- Focus on buffer and K_{sp} problems
- Practice numericals heavily
- Avoid unit mistakes